

Incident Investigation: SSH Brute Force & Privilege Escalation on AWS EC2

Role Demonstrated

Tier 1 SOC Analyst / Incident Responder

Scenario

An AWS EC2 instance (ip-172-31-35-28) was targeted by an external attacker. The auth.log file was provided for analysis. The objective was to determine if a breach occurred, identify the attack vector, and document post-compromise activities.

Tools Used

- Splunk Enterprise (local instance)
- Auth.log (Linux authentication log)
- Manual log correlation and timeline analysis

Investigation Summary

Step 1 – Detection

Using Splunk, I identified a high volume of failed SSH authentication attempts from a single source IP: 65.2.161.68. The failures targeted multiple usernames including admin, backup, server_admin, and root within a short time window, indicating an automated brute force attack.

Step 2 – Confirmation of Breach

A successful root login was recorded at 06:31:40 from the same source IP, confirming the attacker gained unauthorized access. A second successful root login occurred at 06:32:44, followed by an extended session.

Step 3 – Post-Compromise Analysis

The attacker created a new group and user named cyberjunkie, added the account to the sudo group, accessed /etc/shadow, and downloaded a privilege escalation script from GitHub.

Step 4 – Containment Recommendations

- Block source IP 65.2.161.68 at security group level
- Remove malicious user cyberjunkie and associated group

- Revoke and rotate all credentials for root and affected users
- Terminate active sessions from the source IP
- Audit for persistence mechanisms (cron jobs, SSH keys)

Attack Timeline

Time	Activity	Evidence
06:34:18	Created a new group	groupadd
06:34:18	Created a new user	useradd
06:35:15	Added user to sudo group	usermod
06:37:57	Read /etc/shadow file	sudo cat /etc/shadow
06:39:38	Downloaded privilege escalation script	curl to GitHub (linper.sh)

Key Findings

- Attack Vector: SSH brute force
- Attacker IP: 65.2.161.68
- Compromised Users: root, then cyberjunkie
- IOC URL: raw.githubusercontent.com/montysecurity/linper/main/linper.sh

Outcome

The incident was successfully detected through log analysis. A full timeline was reconstructed using Splunk queries. Containment steps were documented and prioritized. This investigation demonstrates proficiency in Splunk search and correlation, Linux authentication log analysis, incident documentation and reporting, and threat hunting for post-exploitation activity.